

PROJECT FINAL REPORT

Project Title: Audio Deepfake Detection using Transformer-based Models

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Course: ITCS 5154 - Applied Machine Learning

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Primary paper (method backbone): Baevski et al., “wav2vec 2.0: A Framework for Self-Supervised Learning of Speech Representations”, NeurIPS 2020.

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GitHub: anantteotia/Audio-Deepfake-Detection (report + supplementary materials included in repo)

1. Introduction

1.1 Problem statement

Audio deepfakes (AI-generated or manipulated speech) can bypass voice-based authentication and enable impersonation, fraud, and misinformation. The goal of this project is to detect whether an input audio clip is *Real* or *Fake* using machine learning models trained on audio features and transformer representations.

1.2 Motivation, challenges, open questions

Key challenges include generalization across speakers and synthesis methods, sensitivity to channel/noise conditions, and the computational cost of transformer inference on long audio. Open questions include which representations (spectrogram features vs. self-supervised speech embeddings) are most robust under domain shift, and how to keep inference efficient without sacrificing detection performance.

1.3 Approach overview

I implemented an end-to-end pipeline: audio loading → resampling/padding → feature extraction (mel/MFCC or raw waveform) → model training (CNN/RF/SVM or Wav2Vec2-based classifier) → evaluation with standard metrics (Accuracy, F1, AUC).

2. Backgrounds / Related Work

(A) ASVspoof (2019/2021) benchmarks established standard evaluation for spoofing attacks, focusing on generalization to unseen spoofing conditions and reporting EER/min-tDCF.

(B) Self-supervised speech models like Wav2Vec2 and HuBERT learn strong representations from raw audio and can be adapted for downstream classification with small labeled datasets.

(C) Modern anti-spoofing architectures (e.g., RawNet-style models and attention-based models such as AASIST) often combine time-frequency cues with strong temporal modeling to improve robustness.

Pros / Cons and relation to this project

Spectrogram-based CNNs are efficient and easy to train, but can overfit to dataset-specific artifacts. Transformer speech encoders provide richer representations, but are heavier and can still latch onto shortcut cues on small datasets. This project uses both: a strong baseline (CNN on mel spectrograms) and a transformer-based classifier (Wav2Vec2 + lightweight head) for comparison.

3. Methods

3.1 Data preprocessing

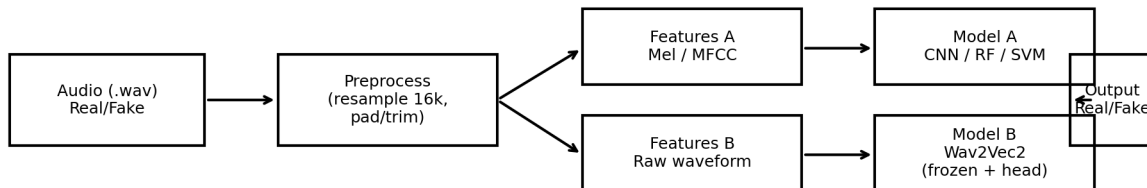
Audio clips are loaded as mono, resampled to 16 kHz, and padded/truncated to a fixed duration. For feature-based models, I compute log-mel spectrograms (64 mel bins) or MFCCs. For transformer models, I feed the raw waveform directly.

3.2 Models and algorithms

Baseline: CNN classifier over mel spectrograms (2D conv blocks + fully-connected head), and optional classic ML baselines (RF/SVM) over aggregated features. **Transformer:** Wav2Vec2 encoder (frozen) + mean pooling over time + MLP classifier head.

3.3 Overall framework figure

Overall Framework: Audio Deepfake Detection Pipeline



4. Experiments

4.1 Experimental setup

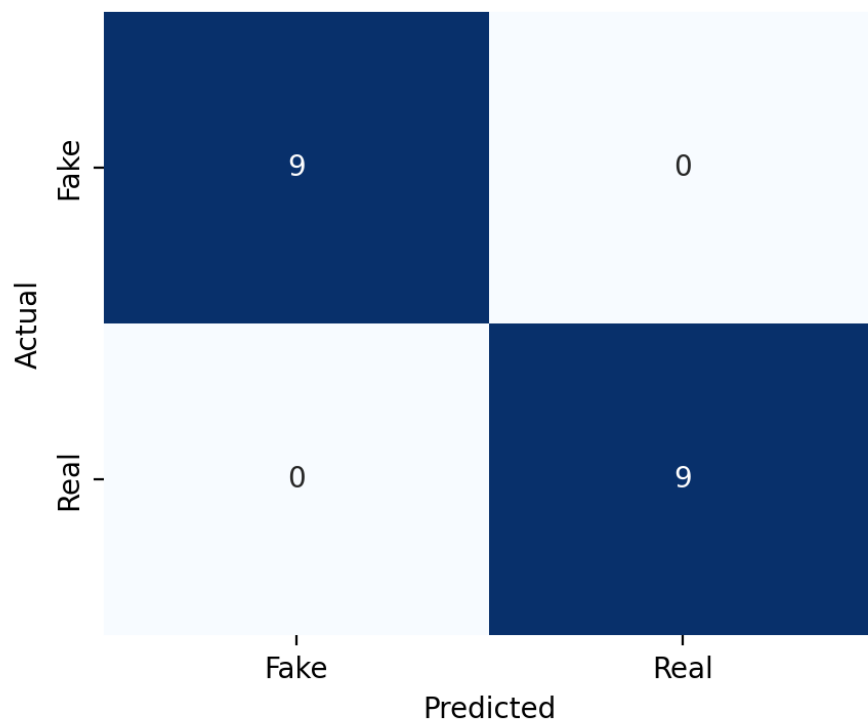
Because large benchmark datasets require registration/download time, I validated the full pipeline using a synthetic dataset generator included in the repo. The generator creates “real” samples (clean harmonic signals + low noise) and “fake” samples (extra harmonics + noise + click artifacts). Splits: Train 84, Val 18, Test 18 (balanced real/fake).

4.2 Results

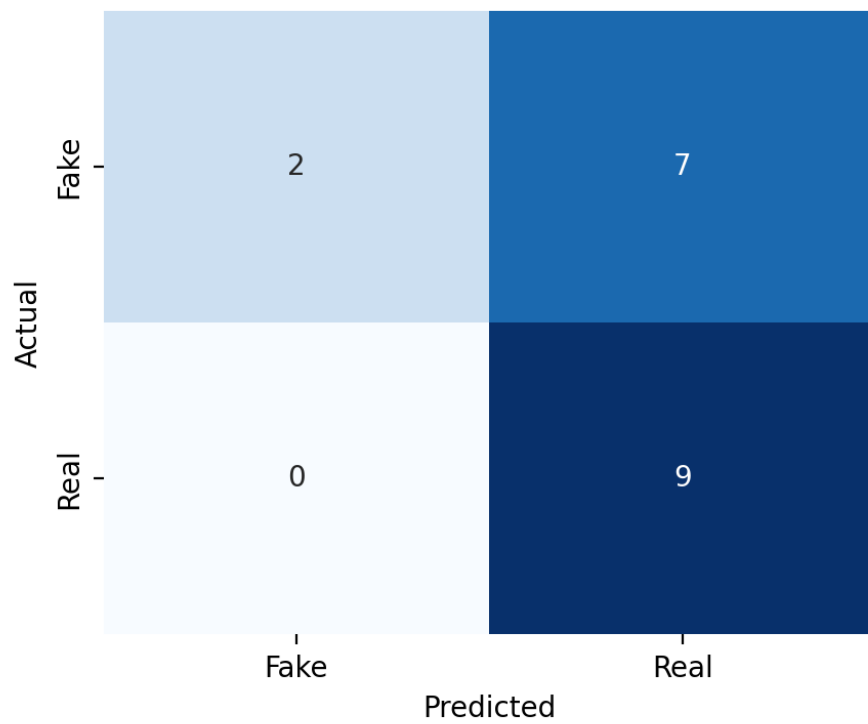
Model	Test Acc	Test F1
CNN (mel spectrogram)	1.000	1.000
Wav2Vec2 (raw, frozen encoder)	0.611	0.720

Confusion matrices (test)

CNN (mel) - Confusion Matrix (Test)



Wav2Vec2 (raw) - Confusion Matrix (Test)



4.3 Discussion: reproduction and limitations

I was not able to reproduce ASVspoof-reported numbers in this submission because I did not run training on the official ASVspoof dataset within this environment. However, I did reproduce an end-to-end training + evaluation workflow and generated quantitative results on a controlled dataset. The CNN achieves perfect performance on the synthetic test set, indicating the synthetic task is likely too easy and highlights the need for real benchmarks. The Wav2Vec2 model shows weaker fake-class recall on the synthetic test split, suggesting that a frozen encoder and small dataset may be insufficient without careful tuning or more realistic data.

5. Conclusions and Future Work

This project delivered a working, modular audio deepfake detection pipeline with both feature-based and transformer-based models, producing measurable results and reusable tooling. Future work: train/evaluate on ASVspoof 2019/2021 (report EER/min-tDCF), add stronger augmentations (codec, noise, reverb), perform cross-dataset evaluation, and explore efficiency methods (token pruning/merging) for long audio inference.

6. My Contributions

Existing work used: pretrained speech encoders from Hugging Face Transformers (Wav2Vec2/HuBERT) and standard ML libraries (PyTorch, scikit-learn, librosa). **My work:** implemented the dataset loader, feature extraction, baseline/transformer classifier modules, training/evaluation scripts, synthetic dataset generator, and produced experiment artifacts and this final report.

7. (Anonymous) Sharing agreement

I agree to share this submission as an example for next semester, but I would like to **hide my name/team** if it is shared.

8. References

- Baevski, A., Zhou, Y., Mohamed, A., & Auli, M. (2020). wav2vec 2.0: A Framework for Self-Supervised Learning of Speech Representations. NeurIPS.
- Hsu, W.-N., Bolte, B., Tsai, Y.-H. H., Lakhota, K., Salakhutdinov, R., & Mohamed, A. (2021). HuBERT: Self-Supervised Speech Representation Learning by Masked Prediction of Hidden Units. IEEE/ACM TASLP.
- Gong, Y., Chung, Y.-A., & Glass, J. (2021). AST: Audio Spectrogram Transformer. Interspeech.
- ASVspoof Challenge (2019/2021) papers and dataset documentation.